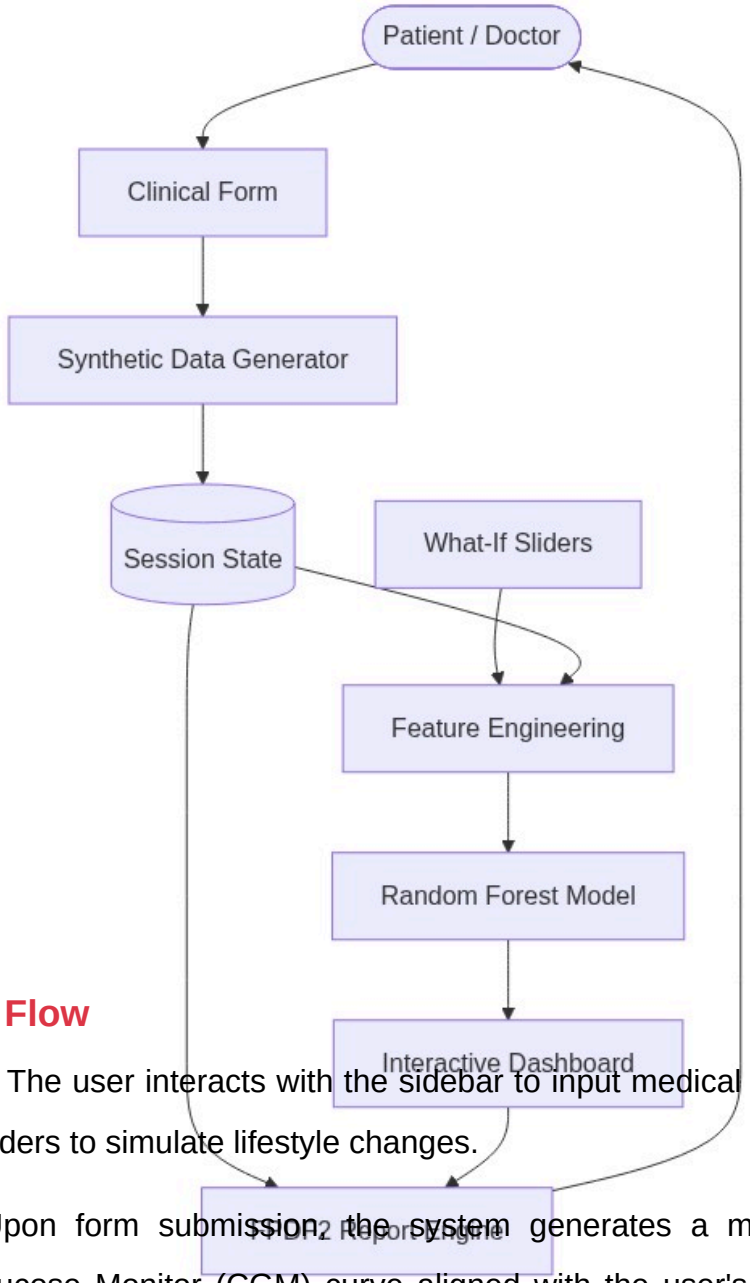
GlucoTwin AI: System Architecture



Data & Processing Flow

1. Frontend (Streamlit): The user interacts with the sidebar to input medical vitals (HbA1c, BP, etc.) and uses the What-If sliders to simulate lifestyle changes.

2. Backend Engine: Upon form submission, the system generates a mathematically synthetic 24-hour Continuous Glucose Monitor (CGM) curve aligned with the user's specific clinical inputs, storing everything in the app's secure Session State.

3. Machine Learning Core: The system extracts time-series features (like rolling averages and lagged glucose changes) and feeds them into a trained Scikit-Learn Random Forest Classifier, which calculates the exact probability of a dangerous glucose spike happening in the next 2 hours.

4. Offline Reporting Engine: The application uses Kaleido to capture headless screenshots of the interactive Plotly graphs, which are dynamically embedded into a highly structured FPDF-generated clinical report downloaded securely by the user.